CS-4347 Airport Management System

Milestone 2 — README

Overview

Command-line Python application connecting to Microsoft SQL Server in Docker, implementing all Milestone 2 functional requirements.

Requirements

Requirement	Version
Python	3.9+
pyodbc	5.0+
ODBC Driver for SQL Server	18
Docker Desktop	Latest

Step 1 — Install Docker Desktop

Download from https://www.docker.com/products/docker-desktop and make sure it is running.

Step 2 — Start SQL Server in Docker

```
docker run -e ACCEPT_EULA=Y -e SA_PASSWORD=StrongPass123 -p 1433:1433 --name sqlserver -d mcr.microsoft.com/m
```

Wait 30 seconds, then create the database:

```
docker exec -it sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost,1433 -U sa -P StrongPass123 -No -Q "CREAT
```

Step 3 — Load the Data

Copy CSVs from Canvas into Docker (replace ~/Downloads with your CSV folder):

```
docker cp ~/Downloads/. sqlserver:/var/opt/mssql/import/
```

Copy and run the Milestone 1 setup script:

```
docker cp setup_m1.sql sqlserver:/var/opt/mssql/import/setup_m1.sql
docker exec -it sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost,1433 -U sa -P StrongPass123 -No -d Airpo
```

Step 4 — Apply Milestone 2 Schema

```
docker exec -it sqlserver /opt/mssql-tools18/bin/sqlcmd -S localhost,1433 -U sa -P StrongPass123 -No -d Airpo
```

Step 5 — Install Python Dependencies

```
pip3 install pyodbc
brew tap microsoft/mssql-release https://github.com/Microsoft/homebrew-mssql-release
brew install msodbcsql18
```

Step 6 — Run the Application

```
python3 airport.py
```

Features

1. Travel Itinerary Search — direct and one-stop flights by city name or airport code
2. Flight Details by Number — all legs, routes, times, and fares
3. Aircraft Utilization Report — every airplane with total flights in a date range
4. Seat Availability Check — total seats vs bookings for a specific flight and date
5. Passenger Itinerary Retrieval — all booked legs for a customer by name

Notes

- SQL Server password: StrongPass123
- SQL Server runs on localhost port 1433 inside Docker
- Docker Desktop must be running before launching the app
- The Milestone 1 CSV files from Canvas are required for data

Third-Party Modules

pyodbc — Python ODBC bridge for SQL Server connectivity